

Pin connection diagram for ESP32-53-WROOM-1U to NI682. The diagram shows a breadboard with two rows of pins. The top row (pins 1-16) is connected to the ESP32: EN (1), TXD0 (2), RXD0 (3), RXD0 (4), TXD0 (5), TXD0 (6), TXD0 (7), TXD0 (8), TXD0 (9), TXD0 (10), TXD0 (11), TXD0 (12), TXD0 (13), TXD0 (14), TXD0 (15), TXD0 (16). The bottom row (pins 17-32) is connected to the NI682: EN (17), TXD0 (18), RXD0 (19), RXD0 (20), TXD0 (21), TXD0 (22), TXD0 (23), TXD0 (24), TXD0 (25), TXD0 (26), TXD0 (27), TXD0 (28), TXD0 (29), TXD0 (30), TXD0 (31), TXD0 (32).

[illegible][illegible][illegible][illegible]

SSR terminals 14/9J: $j_{p1} = SSR_{EN}(\text{board} + \text{v5v, watchdog-gated})$
 $j_{p2} = \text{the switched low side, Board-safe: } H720/\text{hold each} \\
\text{input gate down while the E5372 pins are high impedance.}$
 NOT optional: $j_{p3} = \text{level 1 and level 2 generated: it an opto}$
 $\text{only isolates if the SSR loop is powered off-and-on: an opto}$
 both powers it.

TC1 terminal J3 / TC2 terminal J8: $j_{p1} = K_{+}$, $j_{p2} = K_{-}$ – both
 tied, biased near AGND only through each MAX485's internal
 BIOS terminal. Ungrounded-junction boards required: two
 grounded-junction boards in one kin would loop through it.

Nav switch J6 is panel-mounted; Inputs use ESP32 pull-ups.

J11: $IN1/IN2 \rightarrow IN3 \text{ (lio/gas-flow/spare)} + GND$ – dry contact each
 channel to GND.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted
- do not wire to AC mains. Y1/C26 load caps are an ASSUM
value (unverified C.L), see design.py comment. ADE7953 I2C
address 0x38 collides with PCF8574A.